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power point.

There are two common methods of using the decibel: The first is to express a power gain factor in dB. The second is to determine the power gain factor and convert it into dB, which can all be done in one calculation. The formula for this operation is:

$$G(\text{dB}) = 10 \log(P_{\text{out}}/P_{\text{in}})$$

where 'G' is gain in dB, P_{out} power output from the device and P_{in} power input to the device.

With respect to power gain, 3 dB represents a ratio of two to one or a doubling of power:

$$G(\text{dB}) = 10 \log(P_{\text{out}}/P_{\text{in}}) \\ = 10 \log 2 \\ = 3 \text{ dB} \dots \dots \log_2 = 0.3010$$

-3 dB is the frequency where the output power of the circuit is half the power in the pass band. The 3dB bandwidth of an amplifier, also known as the -3dB bandwidth, is defined as the difference between the lower- and upper-half power points.

A graph illustrating the concept of -3dB bandwidth is shown in Fig. 2, where 3dB bandwidth $B = f_{\text{H}} - f_{\text{L}}$.

Q4. In digital television, what does 4:2:2 signify? 4:2 signifies the aspect ratio but what is the significance of the third number?

Nishant Mittal

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A4. The generic term used to describe the three components that make up the full HDTV video signal is YUV, where 'Y' is the luminance or brightness channel and 'U' and 'V' are colour difference channels. The three channels, Y, U and V, together perform exactly the same job as RGB, but with a smaller bandwidth.

The terms 4:2:2, 4:1:1 or 4:2:0 refer to the colour sampling scheme used in the HDTV signal. To reduce the size or bandwidth of an HDTV signal, three



Panasonic HDTV TX-PR50G20

different sampling schemes are used for composing the amount of colour information transmitted in the signal, without reducing the perceived quality of the image to the viewer. This is possible as human eye is much less sensitive to colour detail than it is to brightness detail. It is done by sampling the colour information (U and V colour difference channels) in the HDTV image less frequently than the luminance or brightness information (the Y channel).

In the 4:2:2 sampling scheme, the two colour difference channels (U and V) are sampled every second horizontal pixel of the image. So the effective horizontal colour resolution is half of the luminance or brightness resolution. The colour difference channels are still sampled on every vertical line of the image. So there is no loss in vertical colour resolution.

Bandwidth requirements for YUV and RGB compare as follows:

1. YUV 4:2:2 requires two-third the bandwidth of RGB (4:4:4)
2. YUV 4:1:1 requires half the bandwidth of RGB (4:4:4)
3. YUV 4:2:0 requires half the bandwidth of RGB (4:4:4)

Answers compiled by Col (Retd) N.C. Pande,

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